$$r(t) = \prod_i \left\langle \hat{D}\left(\lambda_i(t)\right) \right\rangle_{\hat{\rho}_{\mathcal{E}_i}} = \prod_i \left\langle \exp\left[\lambda_i(t)\hat{a}_i^\dagger - \lambda_i^*(t)\hat{a}_i\right] \right\rangle_{\hat{\rho}_{\mathcal{E}_i}}, \tag{5.118}$$

i.e., by the product of the expectation values of the operators $\hat{D}\left(\lambda_i(t)\right)$, see (5.103), for the thermal modes $\hat{\rho}_{\mathcal{E}_i}$ of the environment.

It now turns out that

$$\chi(\lambda_i, t) \equiv \left\langle \exp\left[\lambda_i(t)\hat{a}_i^\dagger - \lambda_i^*(t)\hat{a}_i\right] \right\rangle_{\hat{\rho}_{\mathcal{E}_i}} \tag{5.119}$$